

Istio Traffic Management Experiment Report

1. Introduction

This lab provides hands-on experience with Istio's traffic management capabilities, including request routing, traffic shifting, fault injection, and circuit breaking. The Bookinfo application is used to demonstrate how Istio controls traffic behavior in a microservices architecture.

2. Environment Setup

The Bookinfo application was successfully deployed on Kubernetes with Istio enabled. All services and pods were running correctly.

```
C:\Users\31018\Downloads\istio-1.30.0-alpha.2-win\istio-1.30.0-alpha.2>kubectl apply -f samples/bookinfo/platform/kube/bookinfo.yaml
service/details unchanged
serviceaccount/bookinfo-details unchanged
deployment.apps/details-v1 unchanged
service/ratings unchanged
serviceaccount/bookinfo-ratings unchanged
deployment.apps/ratings-v1 unchanged
service/reviews unchanged
serviceaccount/bookinfo-reviews unchanged
deployment.apps/reviews-v1 unchanged
deployment.apps/reviews-v2 unchanged
deployment.apps/reviews-v3 unchanged
service/productpage unchanged
serviceaccount/bookinfo-productpage unchanged
deployment.apps/productpage-v1 unchanged

C:\Users\31018\Downloads\istio-1.30.0-alpha.2-win\istio-1.30.0-alpha.2>kubectl apply -f samples/bookinfo/networking/bookinfo-gateway.yaml
gateway.networking.istio.io/bookinfo-gateway created
virtualservice.networking.istio.io/bookinfo created

C:\Users\31018\Downloads\istio-1.30.0-alpha.2-win\istio-1.30.0-alpha.2>kubectl apply -f samples/bookinfo/networking/destination-rule-all.yaml
destinationrule.networking.istio.io/productpage created
destinationrule.networking.istio.io/reviews created
destinationrule.networking.istio.io/ratings created
destinationrule.networking.istio.io/details created
```

```
C:\Users\31018>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
bookinfo-gateway-istio-6f4b86fd99-65vtp  1/1     Running   1 (72s ago)  11d
details-v1-764c46cfdb-jfwvj             1/1     Running   1 (72s ago)  11d
productpage-v1-85664dccbc-xqq9x         1/1     Running   1 (72s ago)  11d
ratings-v1-779fdc7f86-qdbpb             1/1     Running   1 (72s ago)  11d
reviews-v1-85964f9f98-95jd9             1/1     Running   1 (72s ago)  11d
reviews-v2-6f7fbd66fd-fzndq             1/1     Running   1 (72s ago)  11d
reviews-v3-689b477554-k2lcj             1/1     Running   1 (72s ago)  11d

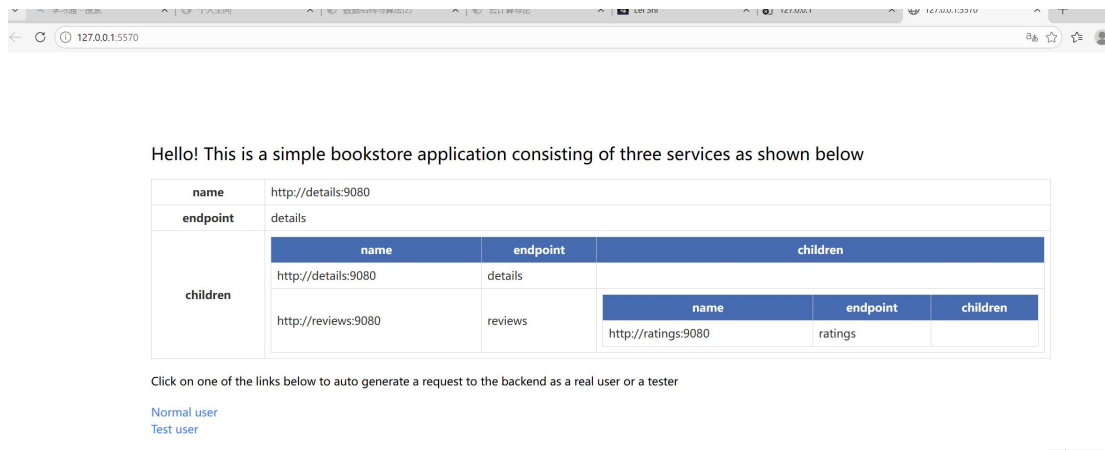
C:\Users\31018>kubectl get svc
NAME                                TYPE           CLUSTER-IP      EXTERNAL-IP      PORT(S)                                AGE
bookinfo-gateway-istio             LoadBalancer  10.96.133.130    127.0.0.1        15021:30355/TCP,80:30763/TCP          11d
details                           ClusterIP      10.104.234.83    <none>           9080/TCP                              11d
kubernetes                         ClusterIP      10.96.0.1        <none>           443/TCP                               11d
productpage                       ClusterIP      10.109.81.54     <none>           9080/TCP                              11d
ratings                           ClusterIP      10.109.67.218    <none>           9080/TCP                              11d
reviews                           ClusterIP      10.110.138.20    <none>           9080/TCP                              11d

C:\Users\31018>kubectl get gateway
NAME                                CLASS  ADDRESS      PROGRAMMED  AGE
bookinfo-gateway                   istio  127.0.0.1     True        11d

C:\Users\31018>kubectl get destinationrules
No resources found in default namespace.
```

Bookinfo Home Page

This is the entry page of the Bookinfo application. It shows the service structure, including details, reviews, and ratings. This confirms that the microservices are correctly deployed and accessible through the Istio Gateway.



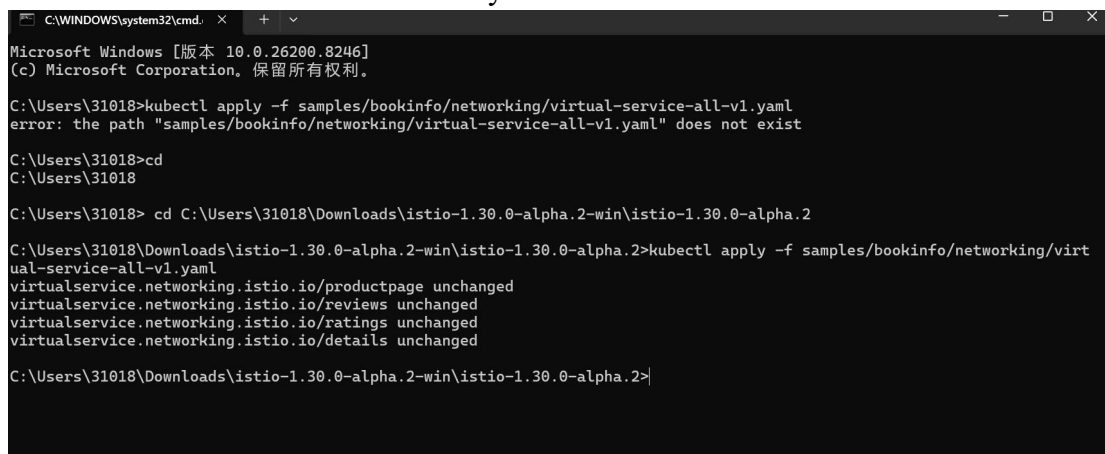
3. Request Routing

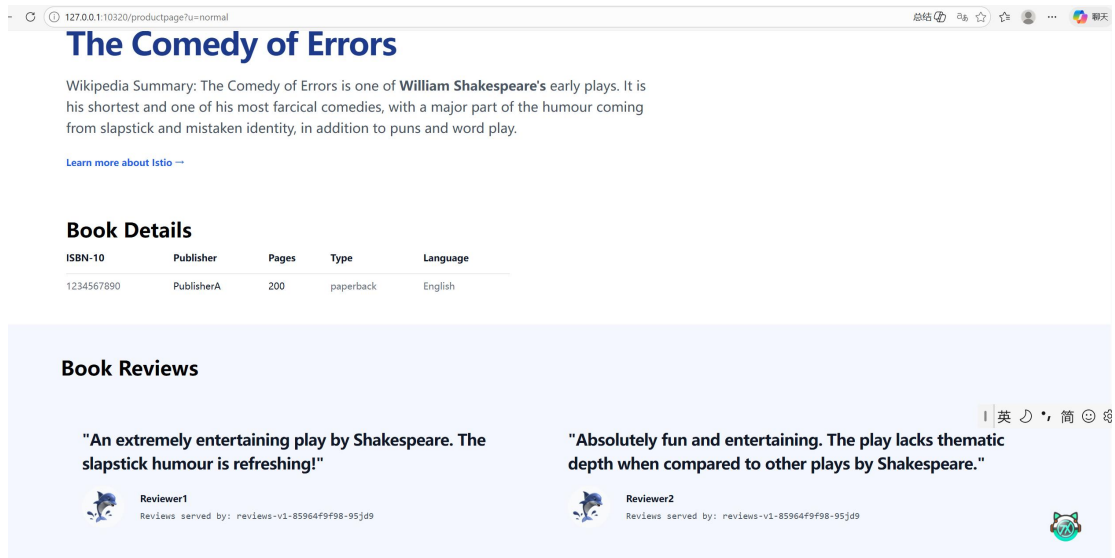
3.1 Route All Traffic to v1

All traffic was first routed to version v1 of the reviews service.

In version v1, the reviews service does not include ratings, so no stars are displayed.

This verifies that traffic is successfully routed to reviews:v1.





3.2 Route User “jason” to reviews:v2

A routing rule was applied so that only user “jason” is directed to reviews:v2.

```
C:\Users\31018\Downloads\istio-1.30.0-alpha.2-win\istio-1.30.0-alpha.2>kubectl apply -f samples/bookinfo/networking/virtual-service-reviews-test-v2.yaml
virtualservice.networking.istio.io/reviews configured

C:\Users\31018\Downloads\istio-1.30.0-alpha.2-win\istio-1.30.0-alpha.2>kubectl get virtualservice reviews -o yaml
apiVersion: networking.istio.io/v1
kind: VirtualService
metadata:
  annotations:
    kubectl.kubernetes.io/last-applied-configuration: |
      {"apiVersion":"networking.istio.io/v1","kind":"VirtualService","metadata":{"annotations":{},"name":"reviews","namespace":"default"},"spec":{"hosts":["reviews"],"http":[{"match":[{"headers":{"end-user":{"exact":"jason"}}}], "route":[{"destination":{"host":"reviews","subset":"v2"}}]},{"route":[{"destination":{"host":"reviews","subset":"v1"}}]}}}
      creationTimestamp: "2026-04-27T06:25:12Z"
    generation: 2
    name: reviews
    namespace: default
    resourceVersion: "41935"
    uid: ac47a7ba-5a88-428c-b8aa-1cc2d498d72e
  spec:
    hosts:
    - reviews
    http:
    - match:
      - end-user:
          exact: jason
      route:
```

Istio uses the HTTP header `end-user` to identify the user. When the user is “jason”, requests are routed to `reviews:v2`, which displays black stars. Other users still access `reviews:v1` without stars.

127.0.0.1:10320/productpage?u=test

BookInfo Sample

Sign out

The Comedy of Errors

Wikipedia Summary: The Comedy of Errors is one of **William Shakespeare's** early plays. It is his shortest and one of his most farcical comedies, with a major part of the humour coming from slapstick and mistaken identity, in addition to puns and word play.

[Learn more about Istio](#)

Book Details

ISBN-10	Publisher	Pages	Type	Language
1234567890	PublisherA	200	paperback	English

Book Reviews

★★★★★

"An extremely entertaining play by Shakespeare. The slapstick humour is refreshing!"

★★★★☆

"Absolutely fun and entertaining. The play lacks thematic depth when compared to other plays by Shakespeare."

127.0.0.1:10320/productpage?u=normal

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Book Details

ISBN-10	Publisher	Pages	Type	Language
1234567890	PublisherA	200	paperback	English

Book Reviews

"An extremely entertaining play by Shakespeare. The slapstick humour is refreshing!"

Reviewer1
Reviews served by: reviews-v1-85964f9f98-95jd9

"Absolutely fun and entertaining. The play lacks thematic depth when compared to other plays by Shakespeare."

Reviewer2
Reviews served by: reviews-v1-85964f9f98-95jd9

4. Traffic Shifting

4.1 50% Traffic to v1 and 50% to v3

Traffic was split evenly between reviews:v1 and reviews:v3.

When refreshing the page multiple times, some requests show no stars (v1), while others show red stars (v3). This demonstrates that Istio distributes traffic based on configured weights.

```
C:\Users\31018\Downloads\istio-1.30.0-alpha.2-win\istio-1.30.0-alpha.2>kubectl get virtualservice reviews -o yaml
apiVersion: networking.istio.io/v1
kind: VirtualService
metadata:
  annotations:
    kubernetes.io/last-applied-configuration: |
      {"apiVersion":"networking.istio.io/v1","kind":"VirtualService","metadata":{"annotations":{},"name":"reviews","name
space":"default"},"spec":{"hosts":["reviews"],"http":[{"route":[{"destination":{"host":"reviews","subset":"v1"},"weight"
:50},{"destination":{"host":"reviews","subset":"v3"},"weight":50}]}]}
creationTimestamp: "2026-04-27T06:25:12Z"
generation: 6
name: reviews
namespace: default
resourceVersion: "43677"
uid: ac47a7ba-5a88-428c-b8aa-1cc2d498d72e
spec:
  hosts:
  - reviews
  http:
  - route:
    - destination:
        host: reviews
        subset: v1
        weight: 50
    - destination:
        host: reviews
        subset: v3
        weight: 50
```

127.0.0.1:10320/productpage?u=normal#

总结 评论 收藏 分享

THE COMEDY OF ERRORS

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ISBN-10	Publisher	Pages	Type	Language
1234567890	PublisherA	200	paperback	English

Book Reviews

★★★★★

"An extremely entertaining play by Shakespeare. The slapstick humour is refreshing!"

 **Reviewer1**
Reviews served by: reviews-v3-689b477554-k21cj

★★★★☆

"Absolutely fun and entertaining. The play lacks thematic depth when compared to other plays by Shakespeare."

 **Reviewer2**
Reviews served by: reviews-v3-689b477554-k21cj

4.2 100% Traffic to v3

All traffic was shifted to version v3.

All requests are now routed to reviews:v3, which displays red stars. This confirms that traffic shifting is working correctly.

```
C:\Users\31018\Downloads\istio-1.30.0-alpha.2-win\istio-1.30.0-alpha.2>kubectl apply -f samples/bookinfo/networking/virtual-service-reviews-v3.yaml
virtualservice.networking.istio.io/reviews configured
```

127.0.0.1:10320/productpage?u=normal#

THE COMEDY OF ERRORS

Wikipedia Summary: The Comedy of Errors is one of William Shakespeare's early plays. It is his shortest and one of his most farcical comedies, with a major part of the humour coming from slapstick and mistaken identity, in addition to puns and word play.

Learn more about Istio

Book Details

ISBN-10	Publisher	Pages	Type	Language
1234567890	PublisherA	200	paperback	English

Book Reviews

★★★★★

"An extremely entertaining play by Shakespeare. The slapstick humour is refreshing!"

Reviewer1
Reviews served by: reviews-v3-689b477554-k21cj

★★★★☆

"Absolutely fun and entertaining. The play lacks thematic depth when compared to other plays by Shakespeare."

Reviewer2
Reviews served by: reviews-v3-689b477554-k21cj

5. Fault Injection

5.1 Delay Injection

A delay fault of 7 seconds was injected into the ratings service for user “jason”.

The page becomes slow or displays an error message such as “product reviews are currently unavailable”. This occurs because the delay exceeds the service timeout, demonstrating how Istio can simulate latency issues.

```
{
  "apiVersion": "networking.istio.io/v1",
  "kind": "VirtualService",
  "metadata": {
    "annotations": {},
    "name": "ratings",
    "namespace": "default"
  },
  "spec": {
    "hosts": ["ratings"],
    "http": [
      {
        "fault": {
          "delay": {
            "fixedDelay": "7s",
            "percentage": {
              "value": 100
            }
          }
        },
        "match": [
          {
            "headers": {
              "end-user": {
                "exact": "jason"
              }
            }
          }
        ],
        "route": [
          {
            "destination": {
              "host": "ratings",
              "subset": "v1"
            }
          }
        ]
      }
    ]
  },
  "creationTimestamp": "2026-04-27T06:25:12Z",
  "generation": 2,
  "name": "ratings",
  "namespace": "default",
  "resourceVersion": "44022",
  "uid": "35016bff-0a83-4628-be26-4d9ce7e6770d"
}
```

5.2 Abort Fault Injection

An HTTP abort fault was injected to simulate service failure.

The ratings service returns an error (HTTP 500), causing the reviews section to fail. This demonstrates how Istio can simulate service failures for testing system resilience.

```
C:\WINDOWS\system32\cmd. x + v
: [{"destination": {"host": "ratings", "subset": "v1"}}]]]]}
creationTimestamp: "2026-04-27T06:25:12Z"
generation: 3
name: ratings
namespace: default
resourceVersion: "44197"
uid: 35016bff-0a83-4628-be26-4d9ce7e6770d
spec:
  hosts:
  - ratings
  http:
  - fault:
      abort:
        httpStatus: 500
        percentage:
          value: 100
      match:
      - headers:
          end-user:
            exact: jason
      route:
      - destination:
          host: ratings
          subset: v1
      - route:
          destination:
            host: ratings
            subset: v1
```

6. Circuit Breaking

6.1 Deploy httpbin Service

The httpbin service was deployed as a test backend.

The httpbin service is used to simulate backend responses for testing circuit breaking behavior.

```
C:\Users\31018\Downloads\istio-1.30.0-alpha.2-win\istio-1.30.0-alpha.2>kubectl apply -f samples/httpbin/httpbin.yaml
serviceaccount/httpbin created
service/httpbin created
deployment.apps/httpbin created

C:\Users\31018\Downloads\istio-1.30.0-alpha.2-win\istio-1.30.0-alpha.2>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
bookinfo-gateway-istio-6f4b86fd99-65vtp  1/1     Running   1 (73m ago)  11d
details-v1-764c46cfd-bjfwj             1/1     Running   1 (73m ago)  11d
httpbin-5695765667-kchcl               0/2     Init:1/2   0          <invalid>
productpage-v1-85664dccbc-xqq9x        1/1     Running   1 (73m ago)  11d
ratings-v1-779fdc7f86-qdbpb            1/1     Running   1 (73m ago)  11d
reviews-v1-85964f9f98-95jd9            1/1     Running   1 (73m ago)  11d
reviews-v2-6f7fbd66fd-fzndq            1/1     Running   1 (73m ago)  11d
reviews-v3-689b477554-k2lcj            1/1     Running   1 (73m ago)  11d

C:\Users\31018\Downloads\istio-1.30.0-alpha.2-win\istio-1.30.0-alpha.2>kubectl get svc httpbin
NAME    TYPE      CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
httpbin ClusterIP  10.110.103.154 <none>        8000/TCP    0s

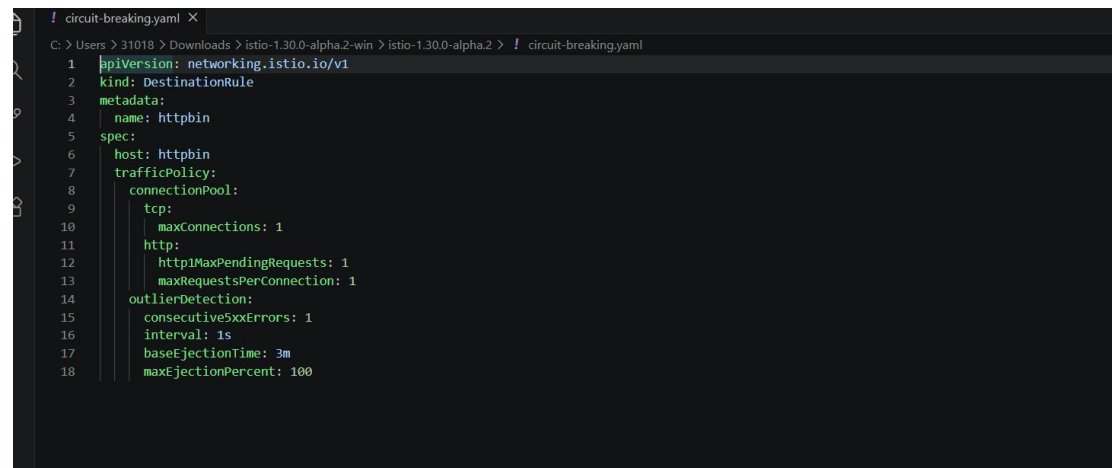
C:\Users\31018\Downloads\istio-1.30.0-alpha.2-win\istio-1.30.0-alpha.2>
```

6.2 Apply Circuit Breaking Rule

A DestinationRule was created to limit connection pools and enable outlier detection.

The rule limits the number of concurrent connections and enables detection of failed requests, forming the basis for circuit breaking.

```
C:\Users\31018\Downloads\istio-1.30.0-alpha.2-win\istio-1.30.0-alpha.2>kubectl apply -f circuit-breaking.yaml
destinationrule.networking.istio.io/httpbin created
```



```
1 apiVersion: networking.istio.io/v1
2 kind: DestinationRule
3 metadata:
4   name: httpbin
5 spec:
6   host: httpbin
7   trafficPolicy:
8     connectionPool:
9       tcp:
10        maxConnections: 1
11      http:
12        http1MaxPendingRequests: 1
13        maxRequestsPerConnection: 1
14      outlierDetection:
15        consecutive5xxErrors: 1
16        interval: 1s
17        baseEjectionTime: 3m
18        maxEjectionPercent: 100
```

```
kind: DestinationRule
metadata:
  annotations:
    kubectl.kubernetes.io/last-applied-configuration: |
      {"apiVersion":"networking.istio.io/v1","kind":"DestinationRule","metadata":{"annotations":{},"name":"httpbin","namespace":"default"},"spec":{"host":"httpbin","trafficPolicy":{"connectionPool":{"http":{"http1MaxPendingRequests":1,"maxRequestsPerConnection":1},"tcp":{"maxConnections":1},"outlierDetection":{"baseEjectionTime":"3m","consecutive5xxErrors":1,"interval":"1s","maxEjectionPercent":100}}}}}
  creationTimestamp: "2026-04-27T07:19:17Z"
  generation: 1
  name: httpbin
  namespace: default
  resourceVersion: "45691"
  uid: 337f8bed-a2fe-426f-89de-d9ad42d62d74
spec:
  host: httpbin
  trafficPolicy:
    connectionPool:
      http:
        http1MaxPendingRequests: 1
        maxRequestsPerConnection: 1
      tcp:
        maxConnections: 1
    outlierDetection:
      baseEjectionTime: 3m
      consecutive5xxErrors: 1
      interval: 1s
      maxEjectionPercent: 100
```

6.3 Generate Traffic with Fortio

High concurrent requests were generated using Fortio.

The result shows both HTTP 200 and HTTP 503 responses. The 503 errors indicate that some requests were rejected due to exceeding the configured limits. This

confirms that circuit breaking was successfully triggered.

```
C:\WINDOWS\system32\cmd. x + v

{
  "args": {},
  "headers": {
    "Host": [
      "httpbin:8000"
    ],
    "User-Agent": [
      "fortio.org/fortio-1.69.5"
    ],
    "X-Envoy-Attempt-Count": [
      "1"
    ],
    "X-Forwarded-Client-Cert": [
      "By=spiffe://cluster.local/ns/default/sa/httpbin;Hash=00282580df9c8714967421367a4824f68e016a58caf64b646e0939a257cf;Subject=\"\";URI=spiffe://cluster.local/ns/default/sa/default"
    ],
    "X-Forwarded-Proto": [
      "http"
    ],
    "X-Request-Id": [
      "f1f14c27-ea61-9af5-bbd5-4a7464c7c2cc"
    ]
  },
  "method": "GET",
  "origin": "127.0.0.6:54117",
  "url": "http://httpbin:8000/get"
}
```

```
C:\Users\31018\Downloads\istio-1.30.0-alpha.2-win\istio-1.30.0-alpha.2>kubectl get pods -l app=fortio
NAME                                READY   STATUS    RESTARTS   AGE
fortio-deploy-5fd895f4b7-z4gzd      2/2     Running   0           2m49s

C:\Users\31018\Downloads\istio-1.30.0-alpha.2-win\istio-1.30.0-alpha.2>kubectl exec -it fortio-deploy-5fd895f4b7-z4gzd f
ortio -- /usr/bin/fortio curl -quiet http://httpbin:8000/get
HTTP/1.1 200 OK
access-control-allow-credentials: true
access-control-allow-origin: *
content-type: application/json; charset=utf-8
date: Mon, 27 Apr 2026 07:24:22 GMT
content-length: 621
x-envoy-upstream-service-time: 32
server: envoy

{
  "args": {},
  "headers": {
    "Host": [
      "httpbin:8000"
    ],
    "User-Agent": [
      "fortio.org/fortio-1.69.5"
    ],
    "X-Envoy-Attempt-Count": [
      "1"
    ]
  },
  "method": "GET",
  "origin": "127.0.0.6:54117",
  "url": "http://httpbin:8000/get"
}
```

circuit breaking was achieved by configuring a DestinationRule to limit connection pools and generating concurrent traffic using fortio. When the request load exceeded the configured limits, some requests returned 503 errors, demonstrating that Istio successfully protected the service from overload.

```
C:\WINDOWS\system32\cmd.exe
> 0.008 <= 0.00857011 , 0.00828505 , 100.00, 1
# target 50% 0.001
# target 75% 0.0035
# target 90% 0.00822804
# target 99% 0.0085359
# target 99.9% 0.00856669
# Socket and IP used for each connection:
[0] 4 socket used, resolved to 10.110.103.154:8000, connection timing : count 4 avg 0.00031255275 +/- 2.083e-05 min 0.000277319 max 0.000331583 sum 0.001250211
[1] 4 socket used, resolved to 10.110.103.154:8000, connection timing : count 4 avg 0.000328819 +/- 6.75e-05 min 0.0002681 max 0.000439558 sum 0.001315276
Connection time histogram (s) : count 8 avg 0.00032068587 +/- 5.061e-05 min 0.0002681 max 0.000439558 sum 0.002565487
# range, mid point, percentile, count
>= 0.0002681 <= 0.000439558 , 0.000353829 , 100.00, 8
# target 50% 0.000341582
# target 75% 0.00039057
# target 90% 0.000419963
# target 99% 0.000437598
# target 99.9% 0.000439362
Sockets used: 8 (for perfect keepalive, would be 2)
Uniform: false, Jitter: false, Catchup allowed: true
IP addresses distribution:
10.110.103.154:8000: 8
Code 200 : 14 (70.0 %)
Code 503 : 6 (30.0 %)
Response Header Sizes : count 20 avg 171.6 +/- 112.3 min 0 max 246 sum 3432
Response Body/Total Sizes : count 20 avg 678.6 +/- 286.5 min 241 max 867 sum 13572
All done 20 calls (plus 0 warmup) 5.644 ms avg, 320.3 qps
```

7. Conclusion

In this experiment, Istio's traffic management capabilities were successfully demonstrated. Request routing allowed traffic control based on user identity. Traffic shifting enabled gradual deployment strategies. Fault injection simulated service delays and failures, helping to test system resilience. Circuit breaking protected the system under high load by rejecting excessive requests with 503 errors. Overall, this lab provided a comprehensive understanding of how Istio manages traffic in microservices environments.